Project Declaration

DIIC · Group 6 · LAB 1 - Tuesday 9:30h - 12:30h

ist1109685 · ist1109366 · ist1109424
ist1110570 · ist1109426

Design context: Colour-blind students - Engineering labs and studies

CO-DESIGNER

Who	Guilherme Quinteiro, 5th-year MSc student in Mechanical Engineering, finishing his thesis.
Colour vision	Mild red-green colour vision deficiency (CVD), deuteranomaly or protanomaly. Hardest pairs: red/green, dark blue/purple, green/brown. Warm light makes it worse.
Where it matters	Mostly in work and studies, above all graphs and diagrams: colour scales in SolidWorks simulations, colour legends in PrusaSlicer (3D printing), graphs with several coloured lines, colour-coded maps. In daily life it mainly affects choosing clothes.
Current strategies	Asking others (frequently), reasoning from values and their order, or intuition. Used every time, but rated as helping only "a little" (2/5). He has never tried a colour filter, app or glasses.
Relationship	Known to the team through Manuel Moiteiro. Takes part voluntarily and may withdraw at any time.

PROBLEM STATEMENT

> *Colour-blind engineering students like Guilherme need to match the colours in technical graphs and simulation colour scales to the values they represent, in order to interpret their own results independently.*

ARGUMENT

About 8% of men and 0.5% of women worldwide have a red-green colour vision deficiency, deuteranomaly alone affecting about 5% of men [1]: at least one student in an average mixed science class of 30 [2]. Engineering relies heavily on colour to carry data, yet rainbow colour maps and red-green pairings remain common, even though they can distort the data shown by over 7% and are unreadable for people with CVD [3]. Colour-only graphs, diagrams and software interfaces are among the main barriers for STEM students with CVD, who "spend their time trying to work out what is being explained" instead of absorbing it [2]. Existing aids fall short: colour-correcting glasses showed no measurable improvement in standard tests [4], and recolouring tools are often designed from simulations rather than lived experience [5].

Guilherme experiences this first-hand. In a static compression test in SolidWorks, he could not tell from the colour scale where the localized forces and displacements were. He has misread results "a few times", typically confusing very different values whose colours look alike, and each colour-coded figure costs him up to a minute of extra checking or a request for help. For colour-blind engineering students like him, every colour-coded result they cannot read alone costs time, risks misinterpretation, and makes their own analysis depend on others.

What the co-designer gains: a voice in shaping a solution to a difficulty in his own daily work, and a prototype he can try and critique.

CO-DESIGNER APPROVAL

I, the undersigned co-designer, confirm that:

- ☐ the description above accurately reflects my experience;
- ☐ I agree with the problem statement above as the focus of this project;
- ☐ I take part voluntarily and may withdraw at any time.

Name: Guilherme Quinteiro	Signature	Date

References

1. Colour Blind Awareness. Types of colour blindness. colourblindawareness.org
2. Institute of Physics (2020). Supporting students in STEM with colour vision deficiency. iop.org
3. Crameri, F., Shephard, G. E., Heron, P. J. (2020). The misuse of colour in science communication. *Nat. Commun.* 11, 5444. nature.com
4. Gómez-Robledo, L. et al. (2018). Do EnChroma glasses improve color vision for colorblind subjects? *Opt. Express* 26(22). optica.org
5. Geddes, C. et al. (2023). 30 Years of Solving the Wrong Problem: How Recolouring Tool Design Fails those with CVD. *ASSETS '23.* dl.acm.org